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Predicting post-stroke pneumonia using deep neural network approaches

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ABSTRACT

Background and purpose: Pneumonia is a common complication after stroke, causing an increased length of hospital stay and death. Therefore, the timely and accurate prediction of post-stroke pneumonia would be highly valuable in clinical practice. Previous pneumonia risk score models were often built on simple statistical methods such as logistic regression. This study aims to investigate post-stroke pneumonia prediction models using more advanced machine learning algorithms, specifically deep learning approaches.

Methods: Using a hospital's electronic health record(EHR) data from 2007-2017, 13,930 eligible patients with acute ischaemic stroke (AIS) were identified to build and evaluate the models (85% of the patients were used for training, and 15% were used for testing). In total, 1012 patients (7.23%) contracted pneumonia during hospitalization. A number of machine learning methods were developed and compared to predict pneumonia in the stroke population in China. In addition to the classic methods (i.e., logistic regression (LR), support vector machines (SVMs), extreme gradient boosting (XGBoost)), methods based on multiple layer perceptron (MLP) neural networks and recurrent neural network (RNNs) (i.e., attention-augmented gated recurrent unit (GRU)) are also implemented to make use of the temporal sequence information in electronic health record (EHR) systems. Prediction models for pneumonia were built for two time windows, i.e., within 7 days and within 14 days after stroke onset. In particular, pneumonia occurring within the 7-day window is considered highly associated with stroke (stroke-associated pneumonia, SAP).

Main findings: The attention-augmented GRU model achieved the best performance based on an area under the receiver operating characteristic curve (AUC) of 0.928 for pneumonia prediction within 7 days and an AUC of 0.905 for pneumonia prediction within 14 days. This method outperformed the other machine learning-based methods and previously published pneumonia risk score models. Considering that pneumonia prediction after stroke requires a high sensitivity to facilitate its prevention at a relatively low cost (i.e., increasing the nursing level), we also compared the prediction performance using other evaluation criteria by setting the sensitivity to 0.90. The attention-augmented GRU achieved the optimal performance, with a specificity of 0.85, a positive predictive value (PPV) of 0.32 and a negative predictive value (NPV) of 0.99 for pneumonia within 7 days and a specificity of 0.82, a PPV of 0.29 and an NPV of 0.99 for pneumonia within 14 days.

Conclusions: The deep learning-based predictive model is feasible for stroke patient management and achieves the optimal performance compared to many classic machine learning methods.

1. Introduction

Stroke is a disease associated with high morbidity, high disability and high mortality worldwide. In China, the number of stroke patients diagnosed each year is approximately 2 million patients, and the mortality rate is 11.48% [1]. Notably, pneumonia is a major cause of death after stroke, especially for acute ischaemic stroke (AIS) patients. It increases the financial burden on medical systems in heavily [2]. Stroke-associated pneumonia (SAP) has an estimated incidence of 5% to 30% [3], as reported by multiple studies from the US [4], Europe [5,6] and China [7,8].

Therefore, it is critical to identify risk factors that are associated with post-stroke pneumonia and to build models that predict pneumonia after AIS. Previous studies have identified a number of risk factors for post-stroke pneumonia, including age [9,10], male sex [11,12], comorbidities such as atrial fibrillation [13] and congestive

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heart failure [14], and stroke severity [15,16]. Different risk models have also been developed to predict post-stroke pneumonia. Kwon et al. [17], Sellars et al. [18] and Hoffmann et al. [10] developed pneumonia scores based on prospective studies, while Chumbler et al. [9] designed a 3-level scoring system for predicting pneumonia in acute stroke patients based on a retrospective study. Key predictors commonly used in these studies include age; dysphagia; standard scores, such as NIHSS or modified Rankin Scale scores; and medical history of pneumonia. For studies on the population of China, Ji et al. [8] developed a 34-point score system using predictors of age; history of atrial fibrillation, congestive heart failure, chronic obstructive pulmonary disease and current smoking; pre-stroke dependence; dysphagia; National Institutes of Health Stroke Scale score at admission, Glasgow Coma Scale score; stroke subtype (Oxfordshire Community Stroke Project) [19]; and blood glucose level.

Most existing post-stroke pneumonia prediction models are built on relatively simple logistic regression models with pre-defined risk factors that are identified by domain experts. Recent advances in data science and artificial intelligence have shown that data-driven approaches that utilize machine learning and deep learning approaches can greatly improve prediction models for medical problems, such as computer-aided diagnosis of breast lesions in ultrasound images and pulmonary nodules in CT scans [20], prediction of Parkinson's disease tremor onset [21], and the Deep Patient project, which uses a deep learning-based unsupervised representation to predict the future of patients from EHRs [22].

Inspired by such new methods, this study plans to develop new pneumonia prediction models using novel machine learning and deep learning algorithms. To the best of our knowledge, this is one of the first studies to apply deep learning to stroke patient management. Once validated, such systems have the potential to detect pneumonia patients after stroke at an early stage, thus guiding appropriate prevention and treatments to avoid severe outcomes.

2. Methods

A set of stroke patients with and without pneumonia was identified from a Chinese hospital's EHRs and divided into two datasets for training and testing. Then, several machine learning methods for pneumonia prediction were implemented and compared: three classic machine learning algorithms (LR, SVM and XGBoost) and two deep learning algorithms (MLP and an RNN-based attention-augmented GRU model). In particular, the temporal sequence information in patient data was used together with other time-insensitive features in the GRU model for a potential performance enhancement.

2.1. Dataset

Stroke patients admitted between 2007 and 2017 were identified from the EHRs of the Second Affiliated Hospital of Nanchang University in China, resulting in a dataset of 13,930 eligible patients, of which 1012 patients had pneumonia (7.23%) during hospitalization after stroke. The pneumonia prediction task is defined to classify whether a patient will have pneumonia after stroke within two time windows: (1) 7 days after stroke (368, 2.64%), which is considered a crucial period for pneumonia highly associated with stroke (i.e., SAP) and (2) 14 days after stroke (738, 5.30%). Therefore, patients in the case group had pneumonia within 7 or 14 days of stroke onset, and patients in the control group were those who did not contract pneumonia within the same time window.

2.1.1. Inclusion criteria of stroke patients

The inclusion criteria for eligible patients were as follows: (1) age > = 18 years; (2) hospitalized with a primary diagnosis of AIS according to World Health Organization criteria [23]; (3) stroke confirmed by head computerized tomography or brain MRI; and (4) direct hospital admission from an outpatient encounter or emergency department. The exclusion criteria were as follows: (1) patients with diseases that have similar clinical manifestations of pneumonia, such as tuberculosis, pulmonary tumour, non-infective interstitial lung disease, pulmonary oedema, pulmonary embolism, and pulmonary atelectasis, were excluded; and (2) only hospital-acquired pneumonia was considered, and patients with pneumonia before the stroke were excluded [24].

2.1.2. Diagnostic criteria for pneumonia

The criteria for pneumonia [25] formulated by the 2010 Chinese Expert Consensus on Diagnosis and Treatment of Stroke-Associated Pneumonia were used, and the detailed clinical criteria for judgement included newly emerging lesions or progressively infiltrating lesions in post-stroke chest images combined with more than two of the following clinical symptoms of infection: (1) fever $\geq 38\,^{\circ}$C; (2) new cough, productive cough, or an exacerbation of pre-existing respiratory disease symptoms with or without chest pain; (3) signs of pulmonary consolidation, and/or moist gales; (4) peripheral WBC $\geq 10 \times 10^9$/L or $\leq 4 \times 10^9$/L with or without nuclear shift to the left.

2.2. Features

2.2.1. Feature description

After a thorough survey of potential risk factors of SAP identified in previous research and a discussion with physicians about the important information available in the hospital EHR system, five types of features are included and evaluated in this study: (1) demographic information, including sex, age group and marital status; (2) vascular risk factors, such as hypertension and diabetes. (3) assessment of nursing levels, such as the score of activities of daily living (ADL) and whether the patient has symptoms of dysphagia; (4) lab test information including the executed lab tests and their frequencies and values within the time windows; (5) medication information including the name and start date of the medication. In total, 1012 lab tests and 1589 medications are used as features. In particular, medications commonly used for SAP treatment were excluded, such as ambroxol hydrochloride. More detailed feature information is illustrated in Table 1.

Table 1
Feature types and instances used for the post-stroke pneumonia predictive modeling.

Feature type	Feature
Demographic information	Age group, gender, marital status
Vascular risk factor	Hypertension, Diabetes mellitus, Dyslipidemia, Atrial fibrillation, Coronary artery disease, History of stroke/TIA, History of surgery
Assessment of nursing level	Conscious disturbance, Dysphagia, Invasive operation, ADL score, tumbling risk factor scale, scale of risk factors for pressure ulcer
Labtest (e.g., Total leukocyte count)	Whether a lab test is taken or not Frequency of a lab test Whether a lab test value is abnormal or not
Medication (e.g., Proton pump inhibitors, thrombolytic therapy)	Whether a medication is taken or not

2.2.2. Feature extraction

Raw clinical data from EHRs were manually checked to remove special symbols, duplicates and obviously erroneous data. Different formats of the same data type, such as time, were converted into a unified format. Numerical values were normalized to reduce the data sparsity: (1) four age groups were created, i.e., < = 59, 60–69, 70–79 and > = 80 following the work in [8]); (2) lab test results were categorized according to the standard normal ranges, i.e., low, normal and high; (3) the ADL scores were normalized to five groups, i.e., normal (= 100), mild (61–99), modest (41–60), severe (< 40) and completely dependent (= 0); (4) the falling risk factor scales were categorized into three groups, i.e., mild (< 25), modest (25–45), and severe (> 45); (5) the scales of the risk factors for pressure ulcers were divided into five groups, i.e., normal (18–23), mild (15–18), modest (13–14), severe (10–12), extremely severe (6–9); and (6) missing values were uniformly considered "unknown".

2.3. Machine learning algorithms

In this work, the performances of different machine learning methods were compared: three classic methods, i.e., LR (logistic regression), SVM (support vector machine), XGBoost (extreme gradient boosting) and two deep neural network methods, i.e., MLP (multi-layer perceptron) and GRU (gated recurrent unit).

Classic machine learning methods

(1) LR: LR estimates the parameters of a logistic model; it is a form of binomial regression. Mathematically, a binary logistic model has a dependent variable with two possible values, such as healthy/sick. In the logistic model, the log-odds (i.e., the logarithm of the odds) of the value labelled "1" is a linear combination of one or more independent variables ("predictors") [26]. For a continuous variable X, in logistic regression, where $F(x) = P(X <= x) = \frac{1}{1+e^{-(x-u)/y}}$, the logistic regression can be calculated as $P(Y = 1|x) = \frac{\exp(w \bullet x)}{1+\exp(w \bullet x)}$.

(2) SVM: SVMs are based on the statistical learning theory of the VC dimension and the structural risk minimization principle [27]. SVMs use kernel functions (e.g., radial basis kernel functions and linear kernel functions) to project high-dimensional samples into lower dimensions to improve the prediction or classification ability of the model. Generally, the essential principle of a linear SVM is to find a hyperplane that can separate data with the maximum margin, which can be formulated as $\min_{w,\, b} \frac{1}{2}||w||^2$

s.t. $y_i(w \bullet x_i + b) - 1 >= 0$, i = 1,2,…,N

where w is the normal vector and b is the intercept of the separation hyperplane.(3) XGBoost: XGBoost is a scalable tree-based gradient boosting algorithm[28]. It generates accurate predictions by integrating weak classifiers. The traditional gradient boosting model uses only the first derivative in the optimization, but XGBoost preforms the second-order Taylor expansion of the cost function and inserts a regularization item into the cost function for better performance. In addition, it learns from a random forest to reduce the calculation and prevent over-fitting through column sampling. The objective function of XGBoost is

$$f = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k)$$

where k is the number of trees, $\sum_{i=1}^{n} l(y_i, \hat{y}_i)$ is the training loss, and $\sum_{k=1}^{K} \Omega(f_k)$ is the model complexity. Specifically,

$$\Omega(f_k) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2$$

where T is the number of leaf nodes, w is the vector of leaf nodes, γ denotes the difficulty degree of splitting nodes and λ is the L2 regularization coefficient.

2.3.1. Deep learning methods

MLP: An MLP-based neural network is a class of feedforward artificial neural networks. It consists of at least three layers of nodes: an input layer, a hidden layer and an output layer. Except for the input nodes, each node is a neuron that uses a non-linear activation function. MLPs utilize a supervised learning technique called backpropagation for training [29,30]. Its multiple layers and non-linear activation function distinguish MLPs from linear perceptrons. It can distinguish data that are not linearly separable [31]. In an MLP, the cross-entropy is used as the loss function: $f = -\frac{1}{n}\Sigma[y \cdot \ln a - (1-y) \cdot \ln(1-a)]$, where y denotes the gold-standard label, a denotes the prediction label, and n denotes the number of labels. Moreover, tanh ($y = (e^x - e^{-x})/(e^x + e^{-x})$) is used as the activation function.

(5) Attention-augmented GRU: Traditional neural networks such as MLPs assume that all inputs (and outputs) are independent of each other, which is not suitable for tasks with sequential information such as time series prediction for clinical outcomes. To address this problem, RNNs perform the same task for every element of a sequence, taking the previous output or hidden states as input (i.e., memory) for the current output prediction [32]. However, the original vanilla RNNs suffer from the vanishing/exploding gradient problem [33]. Therefore, the GRU model, which is a variant of the long short-term memory (LSTM) model, is designed to use a update gate (i.e., a vector) to determine how much of the past information (from previous time steps) needs to be passed to the future and a reset gate to decide how much of the past information to remove [30]. To further improve the interpretability and performance of the model, this study used the architecture of an attention-based GRU, the REverse Time AttentIoN model (RETAIN), which adds neural attention models to a GRU that can detect influential and significant variables in time series data [34].

The original RETAIN experiment used a longitudinal sequence of multiple patient visits to predict heart failure at the next visit. Clinical codes of conditions and medications, which are time-sensitive variables, were used as features in each patient visit. In this study, the main target is to predict whether stroke patients will have pneumonia after hospitalization within a certain time window. However, this study has the same assumption as the original RETAIN work that recent risk factors have higher weights in clinical decision making than distant risk factors. To apply the framework of RETAIN to the data in this study, we transformed our problem by considering each day in the hospital as a "visit", and multiple days form a longitudinal sequence of "visits"within a time window. Furthermore, another difference of this study is that important clinical variables for prediction contain both time-insensitive features, such as age, sex, history of pneumonia and ADL score taken when the patient is enrolled, and time-sensitive features, such as the sequence of medications and the time of abnormal lab test results. Time-insensitive variables were included in the feature set of each day, while time-sensitive variables are present only on the day they occurred.

As illustrated in Fig. 1, given an input sequence $d_1, \ldots, d_i$ (i ∈ [0,7] for the 7-day window, and i ∈ [0,14] for the 14-day window), we predict the label y_i. In the RETAIN architecture, the input layer generates embeddings v1, …, vi of the input sequence $d_1, \ldots, d_i$. Then, two sets of weights are used to represent the sequence information: one set of weights for the day-level attention and the other set of weights for the clinical variable-level attention. The scalars $\alpha_1, \ldots, \alpha_i$ are the day-level attention weights that govern the influence of each day embedding v1, …, vi. An RNN (RNN_α) is used to generate α values:

$$g_i, \ldots, g_1 = RNN_a(v_i, \ldots, v_1)$$

$$e_j = w_a^T g_j + b_a, \text{ for } j = 1, \ldots, i$$

$$a_i, \ldots, a_1 = Softmax_a(e_i, \ldots, e_1)$$

The vectors $b_1, \ldots, b_i$ are the variable-level attention weights that focus on each coordinate of the day embedding $v_{1,1}, v_{1,2}, \ldots, v_{1,m}, \ldots,$

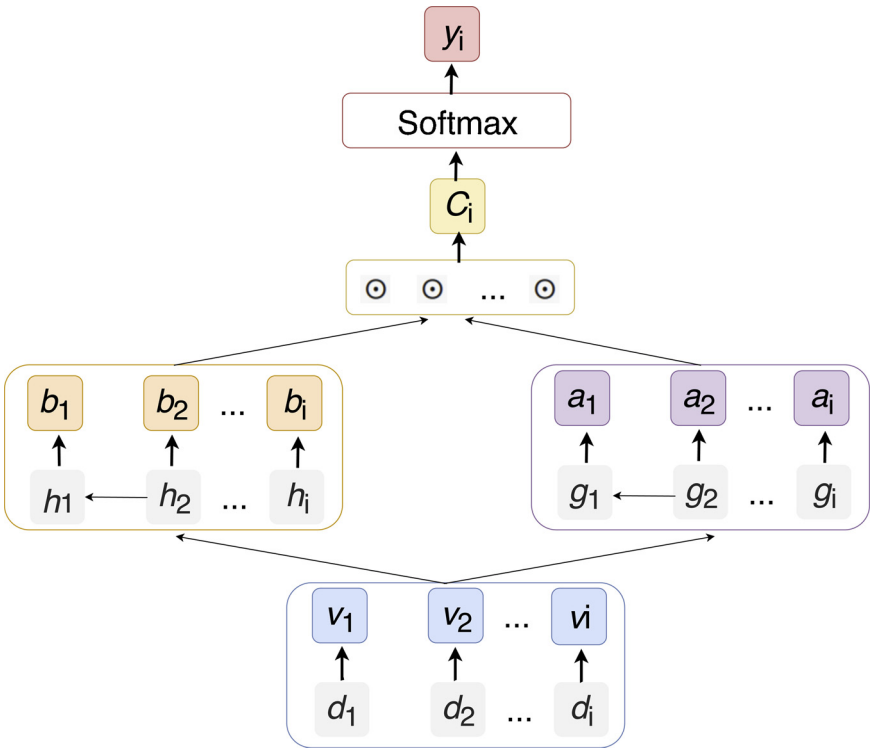

Fig. 1. An illustration of the RETAIN architecture. The temporal sequence is days of hospitalization of stroke patients within time windows of 7 days and 14 days.

$v_{i,1}, v_{i,2}, \ldots, v_{i,m}$, where m is the size of the embedding dimension. An RNN (RNN_β) to generate b values:

$$h_i, \ldots, h_1 = RNN_\beta(v_i, \ldots, v_1)$$

$$b_j = \tanh(w_\beta^T h_j + b_\beta) \text{ for } j = 1, \ldots, i$$

The attention and representation vectors from RNN_α and RNN_β are integrated to generate the context vector C_i,

$$c_i = \sum_{j=1}^{i} a_j \beta_j \odot v_j$$

and to predict the label using the softmax function

$$\hat{y}_i = softmax(Wc_i + b)$$

In particular, the RNNs used the reversed time, i.e., from 7/14 days to the first day of hospitalization. The reversed time used in our framework is motivated by doctors' regular practice, which focuses on important clinical information from the present to the past. This framework assumes that recent risk factors should have a higher weight in clinical decision making than distant risk factors.

2.4. Experiment and evaluation

For each prediction task (7-day prediction vs. 14-day prediction), the dataset was divided into a training set (85%) and a test set (15%). Different machine learning models were built using the training set, and the parameters were continuously adjusted using ten-fold cross-validation to optimize the models. The final performance of each machine learning algorithm was evaluated using the intended test dataset. Standard metrics, i.e., the area under the receiver operating characteristic curve (AUC), sensitivity, specificity, positive predictive value (PPV) and negative predictive value (NPV), were reported to evaluate the prediction performance of different models. The AUC usually provides an overly optimistic view of the performance, but the sensitivity and specificity can provide a more objective model assessment from other aspects. Considering that pneumonia prediction after stroke requires a high sensitivity and to facilitate the prevention of pneumonia at a relatively low cost, we compared the prediction performances using other evaluation criteria by setting the sensitivity to 0.90. Moreover, the proportion between positive and negative instances in the training dataset was adjusted to 1:7 in our experiments to achieve the optimal AUC while balancing the sensitivity and specificity.

In our experiments, the LR and SVM machine learning algorithms were implemented using the scikit-learn version 0.18.0 packages. The XGBoost algorithm was implemented using the XGBoost version 0.72 packages, and the DNN was implemented using TFLearn (a high Application Programming Interface (API) for TensorFlow). The attention-based GRU model was implemented in Theano 0.8. For the SVM model based on the RBF kernel, we selected the best parameters: a C of 10 and a gamma of 0.001. Compared with other models, XGBoost has more parameters to train. The model contained 700 decision trees with a learning rate of 0.01. The maximum depth of the trees is 7, and the minimum sum of the leaf node weights (min_child_weight) is 1. The ratio of randomly sampling per tree (subsample) is 0.8, the proportion of the number of attributes randomly sampled per tree (colsample_by-tree) is 0.8, gamma is 0, and the regularization parameter (reg_alpha) is 0.001. The architecture of our MLP model is composed of three fully connected layers. The first two layers used TanHyperbolic (tanh) as the activation function, and the last layer used the softmax activation function. Sixty-four neurons were used in the first layer, 32 neurons were used in the second layer, and 2 neurons were used in the third layer. The attention-based GRU model, i.e., RETAIN, is composed of an input layer, a hidden layer for clinical variables, a hidden layer of variables from different days, and an output layer. The input embeddings were randomly initiated. Both the input and hidden layers were set to 200 neurons. The dropout rate of each layer is 0.5, and the batch size was set to 32. The L2 regularization coefficient (0.0001) was only applied to the input layer. The hidden layer for clinical variables used tanh as the activation function, and the hidden layer for each day used softmax as the activation function. The logistic sigmoid function was used in the last layer.

Table 2
Data statistics of training and test datasets.

SAP Label	Train	Test	Total
7 days			
Positive	313 (85%)	55 (15%)	368
Negative	11423 (85%)	2139 (15%)	13,562
14 days			
Positive	627 (85%)	111 (15%)	738
Negative	11321 (85%)	1871 (15%)	13,192

Table 3
Comparison of five models for post-stroke pneumonia prediction on the training dataset using 10-fold cross validation.

Method	AUC	Sensitivity	Specificity	PPV	NPV
7 days					
LR	0.894	0.90	0.66	0.30	0.99
SVM	0.827	0.90	0.64	0.29	0.98
XGBoost	0.910	0.90	0.67	0.31	0.98
MLP	0.950	0.90	0.73	0.30	0.98
RETAIN	0.960	0.90	0.91	0.35	0.98
14 days					
LR	0.846	0.90	0.70	0.31	0.98
SVM	0.825	0.90	0.63	0.30	0.94
XGBoost	0.874	0.90	0.74	0.35	0.98
MLP	0.901	0.90	0.76	0.33	0.97
RETAIN	0.905	0.90	0.78	0.39	0.98

3. Results

Table 2 lists the number of positive and negative samples in each dataset (i.e., the training and test dataset) used in this study to build the prediction models for pneumonia within seven days and within fourteen days. In total, among all 13,930 stroke patients, 368 (2.64%) patients had pneumonia (i.e., SAP) within seven days, and 738 (5.26%) patients had pneumonia within fourteen days.

Tables 3 and 4 illustrate the prediction performance of each model on the training dataset based on ten-fold cross-validation and the test dataset, respectively. From the AUC value (Fig. 2,3), the attribute-augmented GRU model (i.e., RETAIN) achieved the best prediction on the test dataset, with an AUC value of 0.928 for the 7-day pneumonia predictions and 0.905 for the 14-day pneumonia predictions. It seems that the deep learning models did achieve better performance. With the same sensitivity, the specificity and PPV are much higher than for other models. The XGBoost models (7-day AUC: 0.887; 14-day AUC: 0.894) achieved the highest AUC among the classic machine learning methods and outperformed the basic DNN-based MLP model. In contrast, the SVM achieved the lowest AUCs among the five methods.

The characteristics of the RETAIN model is that each instance is different, and it changes dynamically over time. We evaluated the interpretability of the RETAIN model in the post-stroke pneumonia prediction task by choosing a pneumonia patient from the test set and calculating the contribution of the variables (in this case, medical codes) to diagnostic prediction (Fig. 4).

Table 4
Comparison of five models for post-stroke pneumonia prediction on the test dataset.

Method	AUC	Sensitivity	Specificity	PPV	NPV
7 days					
LR	0.861	0.90	0.63	0.06	1.00
SVM	0.844	0.90	0.61	0.06	1.00
XGBoost	0.887	0.90	0.71	0.08	1.00
MLP	0.880	0.90	0.64	0.07	1.00
RETAIN	0.928	0.90	0.85	0.32	0.99
14 days					
LR	0.868	0.90	0.67	0.13	0.99
SVM	0.862	0.90	0.64	0.13	0.99
XGBoost	0.894	0.90	0.71	0.15	0.99
MLP	0.871	0.90	0.58	0.11	0.99
RETAIN	0.905	0.90	0.82	0.29	0.99

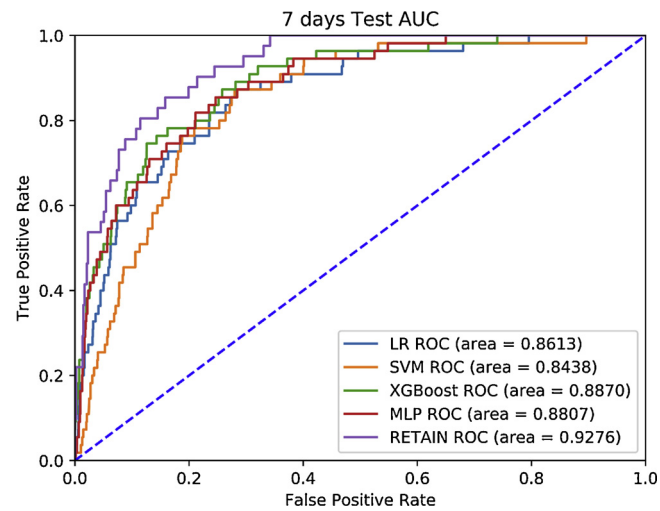

Fig. 2. Receiver operating characteristic curves for the 7-day test datasets.

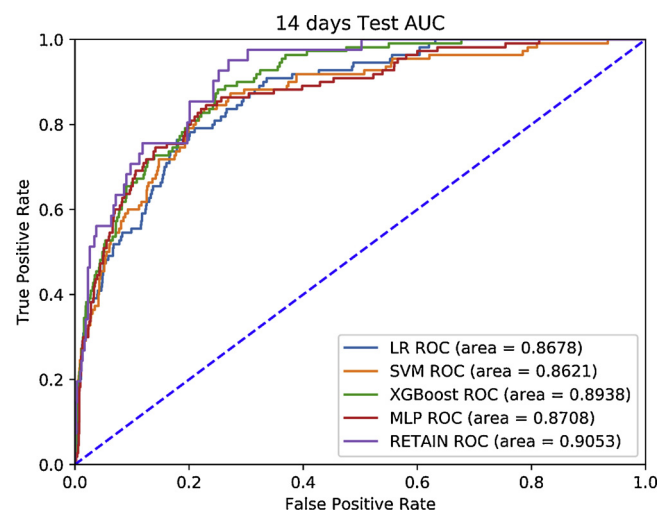

Fig. 3. Receiver operating characteristic curves for the 14-day test datasets.

4. Discussion

Pneumonia is a common complication after AIS, and many research efforts are made to facilitate its prediction. Although multiple scores have been developed for SAP prediction, they are mainly based on a small set of risk factors pre-defined by domain experts that are weighted manually or with the assistance of classic statistical models such as logistic regression. Moreover, such models can only detect high-risk patients, and they are not able to provide a clear time course of pneumonia after AIS [8]. The major reasons are that such models cannot leverage useful information in the full range of the patient data, such as medications and lab test results. Furthermore, the longitudinal characteristics of patient records have not been considered in these static scores for a dynamic prediction within different time windows. This study takes the initiative to develop new post-stroke pneumonia prediction models using novel deep learning algorithms, which combine time-insensitive features such as disease history and demographic information with the time series of medications and lab tests for pneumonia prediction. Our experiments show that the deep learning

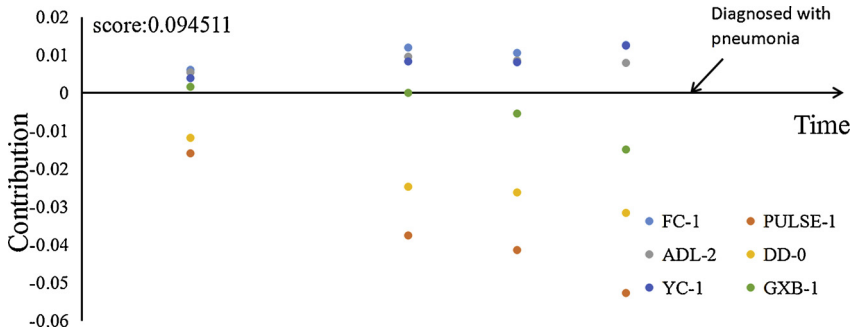

Fig. 4. Features of post-stroke pneumonia along the timeline within the 7-day window.

Table 5
Top weighted features for the prediction of post-stroke pneumonia ranked by the co-efficient of logistic regression.

Rank	Feature	Type
1	Proton pump inhibitors	Medication
2	Heparin sodiumi	Medication
3	Rh (D) blood group	
4	Lactate dehydrogenase	Lab test
5	Ferritin	Lab test
6	Torsaemide for injection	Medication
7	Risk factors for pressure ulcer	Assessment
8	Neutrophil percentage	Lab test
9	Hypertension history	Disease history
10	Disturbance of consciousness	Condition
11	Determination of erythrocyte sedimentation rate (ESR)	Lab test
12	Invasive operation	Procedure
13	Injectio mannitou	Medication
14	Age group	Demographic info
15	Alanine glutamine injection	Medication
16	Omeprazole sodium injection	Medication
17	Hypersensitive cardiac troponin	Lab test
18	Donepezil hydrochloride tablets	Medication
19	C reactive protein (scattering turbidimetry)	Lab test
20	Sodium nitroprusside for injection	Medication
21	Short peptide enteral nutrition liquid	Medication
22	Dysphagia	Condition
23	Furosemide injection	Medication
24	Type B brain natriuretic peptide	Lab test
25	Morse fall assessment form	Assessment

approaches achieved optimal performance when compared with traditional machine learning algorithms, indicating the potential of deep learning in pneumonia patient management.

To further interpret the prediction models, the top features measured by the LR models for SAP (i.e., within the 7-day window) were examined. As illustrated in Table 5, the top 25 features include clinical concepts of conditions, assessments, procedures, disease history, demographic information, lab tests and medications. The information contributed by these features can be categorized into four major types: (1) conditions, demographic information, assessment scores and procedures indicating the nursing levels needed by the patients, such as conditions dysphagia and disturbance of consciousness, age group, the risk factor scores for pressure ulcers, Morse Fall Scale and the invasive operations (e.g., tracheal intubation, tracheotomy, sputum nursing, and ventilator use); (2) lab tests and medications indicating comorbidities (e.g., lactate dehydrogenase is used for the diagnosis of cardiomyopathies, B-type natriuretic peptide and torsemide are used for the indication and treatment of congestive heart failure); (3) lab tests indicating inflammation such as the neutrophil percentage, erythrocyte sedimentation rate (ESR) and C-reactive protein level; (4) clinical variables related to AIS, such as the risk factor for hypertension history, and medications, such as heparin sodium injection and mannitol injection.

Compared with previous work related to the cause analysis and prediction of pneumonia, Yuan et al. analysed the influencing factors of stroke complicated with pulmonary infection and proposed that age, heart disease, atrial fibrillation, chronic obstructive pulmonary disease and dysphagia all have important relationships with lung infections in stroke patients [35]. Risk factors such as age, dysphagia and heart disease were also found in our study, consistent with their results. In terms of pneumonia prediction, the prediction models in our study achieved higher AUCs than previous work using manually designed scores [8–10,12], although these models may not be compared directly, as different cohorts and different features were used in these studies. For example, features not commonly collected in clinical practice were used in previous studies (e.g., NIHSS scores [36]), while longitudinal EHR data were used in our study. Furthermore, the aim of previous studies was to detect high-risk patients, whereas our study built prediction models for pneumonia patients within specific time windows after the onset of stroke. Nevertheless, the high performance of the prediction models validates the hypothesis of this study to use longitudinal EHR data and machine learning methods, specifically deep learning methods, for post-stroke pneumonia prediction in practical clinical settings.

Among the three classic machine learning models, XGBoost achieved the best performance both in terms of the combined predictive performance and the predictive stability. XGBoost, which is an upgrade of gradient boosting, prevents the model from falling into the local optimal solution through pruning. Regularization was used to prevent over-fitting, so its AUC, sensitivity and specificity were relatively high. On the test dataset, RETAIN—the attention-augmented GRU model—achieved the best performance among these models, especially. One potential reason is that the RETAIN model considered both the time-insensitive features and time series of medications and lab test results. In addition, the attention mechanism further highlights important information from each single feature as well as from each day as a whole. Although the number of patients who contract pneumonia within 14 days is more than that within 7 days after stroke onset, the prediction performance for pneumonia within 14 days is lower than that within 7 days. One potential reason is that data become noisier and sparser as the time windows used for prediction increase. Further efforts would be spent on mining frequent patterns from time series or making the models more tolerant to pattern variations to reduce the data sparseness.

Currently, longitudinal risk factors such as lab tests used in our prediction models mainly occur within the two time windows, i.e., 7 days post-stroke and 14 days post-stroke. As explained in the methods section, the effects of the risk factors may decrease along the timeline, which follows the intuition of doctors during the diagnosis process. We explored the performances for each year and found that they were not exactly the same because performances are dependent on the number of patients with stroke and the proportion between patients with and without post-stroke pneumonia. Therefore, the performance in each year varies. However, the deep learning methods consistently

Table 6
AUC of each after-stroke pneumonia prediction model for patients in 2017 on the training dataset using 10-fold cross validation.

AUC	LR	SVM	XGBoost	MLP	RETAIN
7 days	0.632	0.376	0.608	0.940	0.945
14 days	0.492	0.432	0.479	0.884	0.891

outperformed the other methods, as illustrated by the AUC from 2017 as an example in Table 6 (7 days: 1323 stroke patients and 72 patients with pneumonia; 14 days: 1323 stroke patients and 171 patients with pneumonia).

One important and practical issue of the deep learning-based prediction model is whether it can be scalable to larger datasets. For a detailed examination, the computer specifications for the method implementation are listed in Table 7. Traditional machine learning methods LR, SVM, XGBoost, and the basic deep learning algorithm MLP were implemented in a CPU environment, while the RETAIN algorithm was implemented in a GPU environment. Given the complexity of deep learning architecture, the RETAIN model demonstrates its efficiency by requiring ~153 s for training and ~15 s for testing within the 7-day window, and ~287 s for training and ~27 s for testing within the 14-day window when implemented on a GPU sever. Therefore, applying a RETAIN model on a larger dataset is promising. As noted in the discussion section, one limitation of the current study is that only one hospital is involved. If generalizing the model to a nationwide level, the computing speed of re-training the model could be further reduced by using more powerful GPUs and graphics cards.

This study has several limitations. First, this study only used the data of patients with AIS from one hospital, and no external validation was conducted. The models built in this study will be applied to other sites for further investigation of its generalizability. For example, a good validation of the prediction models could be built in this study if it could be applied to the multi-site data collected by Ji et al. [8]. Second, since this is a retrospective study, some important risk factors, such as the NIHSS scores, are not used as in prospective studies. In contrast, this retrospective study applied multiple machine learning-based methods to the big longitudinal data from EHRs in a more scalable, data-driven way. Furthermore, other types of risk factors, such as the classification of AIS subtypes (from the OCSP), have not yet been examined in our current model. Additional risk factors will be extracted and used as features in the future to further improve the performance.

5. Conclusion

This study takes the initiative to develop new pneumonia prediction models for patients with AIS using novel deep learning algorithms, which combine time-insensitive features, such as disease history and demographic information, with time series of medications and lab tests for pneumonia prediction within different time windows. The experimental comparison with classic machine learning methods demonstrated that it is feasible to apply deep learning-based prediction models to stroke patient management, which achieve optimal performance compared with multiple classic machine learning methods.

Summary table

What was already known

- Most existing post-stroke pneumonia prediction models are based on a large number of sample statistics, are analysed by the law of disease occurrence, and artificially formulate evaluation scores for disease prediction.
- With the rapid development of machine learning technology for big data, research on disease prediction models based on medical big data is being increasingly applied in the medical research field, and research on machine learning models in disease prediction is being gradually applied.

What this study added to our knowledge

- This study takes the initiative to develop new post-stroke pneumonia prediction models using longitudinal EHR data and machine learning methods, specifically deep learning methods.
- Compared with traditional machine learning algorithms, the deep learning method achieves the optimal performance, indicating the prospect of using deep learning in post-stroke pneumonia patient management.

Authors' contributions

Conceived and designed this study: Yanqiu Ge, Qinghua Wang

Analyzed and explained the data: Li Wang, Honghu Wu, Yuan Xu, Yaoyun Zhang

Collected and helped with analyzing the data of the study: Chen Peng,Jiajing Wang, Gang Xiong

All of the authors contributed to the writing and approved the final manuscript.

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Declaration of Competing Interest

None.

Table 7
Computer specification and running time of five methods. Running times of both training based on 10-fold cross validation and open test are reported. (Second).

	Computer specification	Days	Training time	Test time
LR	CPU : Intel(R) Xeon(R) CPU E3-1231 v3 @ 3.40GHz	7	4.988735198974609	0.5495545864105225
	Graphics card: NVIDIA Corporation GT218 [GeForce 210]	14	12.427517890930176	1.4480609893798828
SVM	Memory: 16 Gigabytes	7	2749.4151618480682	380.6694347858429
	OS : ubuntu 16.04LTS	14	4517.106971025467	613.3069655895233
XGBoost		7	413.06896471977234	40.19403672218323
		14	2958.397967338562	331.1072111129761
MLP		7	337.15784215927124	38.441304206848145
		14	892.1631457805634	81.37669372558594
RETAIN	Graphics card : NVIDIA Tesla K40 900-22081-2250-000	7	153.3097672178185	15.168670070401476
	Memory: 16 Gigabytes	14	287.49619500402294	27.757234079959954
	OS : GNU/Linux			

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.ijmedinf.2019.103986.

References

[1] Z. Chen, et al., Mortality of stroke and its subtypes in China: results from a nationwide population-based survey, Neuroepidemiology 48 (2017) 95–102.
[2] H.-C. Koennecke, et al., Factors influencing in-hospital mortality and morbidity in patients treated on a stroke unit, Neurology 77 (2011) 965–972.
[3] W.F. Westendorp, P.J. Nederkoorn, J.-D. Vermeij, M.G. Dijkgraaf, D. van de Beek, Post-stroke infection: a systematic review and meta-analysis, BMC Neurol. 11 (2011) 110.
[4] I.L. Katzan, R.D. Cebul, S.H. Husak, N.V. Dawson, D.W. Baker, The effect of pneumonia on mortality among patients hospitalized for acute stroke, Neurology 60 (2003) 620–625.
[5] L.P. Kammersgaard, et al., Early infection and prognosis after acute stroke: the Copenhagen Stroke Study, J. Stroke Cerebrovasc. Dis. 10 (2001) 217–221.
[6] F.H. Vermeij, et al., Stroke-associated infection is an independent risk factor for poor outcome after acute ischemic stroke: data from the Netherlands Stroke Survey, Cerebrovasc. Dis. 27 (2009) 465–471.
[7] P.L. Wang, et al., Effect of in-hospital medical complications on case fatality post-acute ischemic stroke: data from the China National Stroke Registry, Chin. Med. J. (Engl.) 125 (2012) 2449–2454.
[8] R. Ji, et al., Novel risk score to predict pneumonia after acute ischemic stroke, Stroke 44 (2013) 1303–1309.
[9] N.R. Chumbler, et al., Derivation and validation of a clinical system for predicting pneumonia in acute stroke, Neuroepidemiology 34 (2010) 193–199.
[10] S. Hoffmann, et al., Development of a clinical score (A2DS2) to predict pneumonia in acute ischemic stroke, Stroke 43 (2012) 2617–2623.
[11] S. Aslanyan, et al., Pneumonia and urinary tract infection after acute ischaemic stroke: a tertiary analysis of the GAIN International trial, Eur. J. Neurol. 11 (2004) 49–53.
[12] H.-M. Kwon, S.-W. Jeong, S.-H. Lee, B.-W. Yoon, The pneumonia score: a simple grading scale for prediction of pneumonia after acute stroke, Am. J. Infect. Control 34 (2006) 64–68.
[13] B. Ovbiagele, N.K. Hills, J.L. Saver, S.C. Johnston, C.A.S.P.R. Investigators, Frequency and determinants of pneumonia and urinary tract infection during stroke hospitalization, J. Stroke Cerebrovasc. Dis. 15 (2006) 209–213.
[14] H.C. Emsley, S.J. Hopkins, Acute ischaemic stroke and infection: recent and emerging concepts, Lancet Neurol. 7 (2008) 341–353.
[15] U. Walter, et al., Predictors of pneumonia in acute stroke patients admitted to a neurological intensive care unit, J. Neurol. 254 (1323) (2007).
[16] M. Vargas, et al., Clinical consequences of infection in patients with acute stroke: is it prime time for further antibiotic trials? Stroke 37 (2006) 461–465.
[17] H.-M. Kwon, S.-W. Jeong, S.-H. Lee, B.-W. Yoon, The pneumonia score: a simple grading scale for prediction of pneumonia after acute stroke, Am. J. Infect. Control 34 (2006) 64–68.
[18] C. Sellars, et al., Risk factors for chest infection in acute stroke: a prospective cohort study, Stroke 38 (2007) 2284–2291.
[19] P. Amarenco, J. Bogousslavsky, L.R. Caplan, G.A. Donnan, M.G. Hennerici, Classification of stroke subtypes, Cerebrovasc. Dis. 27 (2009) 493–501.
[20] J.-Z. Cheng, et al., Computer-aided diagnosis with deep learning architecture: applications to breast lesions in US images and pulmonary nodules in CT scans, Sci. Rep. 6 (2016) 24454.
[21] D. Wu, et al., Prediction of Parkinson's disease tremor onset using a radial basis function neural network based on particle swarm optimization, Int. J. Neural Syst. 20 (2010) 109–116.
[22] R. Miotto, L. Li, B.A. Kidd, J.T. Dudley, Deep patient: an unsupervised representation to predict the future of patients from the electronic health records, Sci. Rep. 6 (2016) 26094.
[23] W.H. Organization, Task force on stroke and other cerebrovascular disorders: recommendations on stroke prevention, diagnosis, and therapy, Stroke 20 (1989) 1407–1431.
[24] L. Li, L. Zhang, W. Xu, J. Hu, Risk assessment of ischemic stroke associated pneumonia, World J. Emerg. Med. 5 (2014) 209.
[25] Chinese Expert Consensus Group on Diagnosis and Treatment of Stroke-associated Pneumonia, Chinese expert consensus on diagnosis and treatment of stroke-associated pneumonia, Chin J Intern Med 49 (2010) 1075–1077.
[26] S.H. Walker, D.B. Duncan, Estimation of the probability of an event as a function of several independent variables, Biometrika 54 (1967) 167–179.
[27] M.A. Hearst, S.T. Dumais, E. Osuna, J. Platt, B. Scholkopf, Support vector machines, IEEE Intell. Syst. Their Appl. 13 (1998) 18–28.
[28] T. Chen, C. Guestrin, Xgboost: a scalable tree boosting system, Proceedings of the 22Nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, ACM, 2016, pp. 785–794.
[29] C.Van Der Malsburg, Frank rosenblatt: principles of neurodynamics: perceptrons and the theory of brain mechanisms, Brain Theory, Springer, Berlin Heidelberg, 1986.
[30] K. Cho, B. Van Merrienboer, C. Gulcehre, et al., Learning phrase representations using RNN encoder-decoder for statistical machine translation, Comput. Sci. (2014).
[31] G. Cybenko, Approximation by superpositions of a sigmoidal function, Math. Control Signals Syst. 2 (1989) 303–314.
[32] B.G. Horne, C.L. Giles, An experimental comparison of recurrent neural networks, Adv. Neural Inf. Process. Syst. 7 (1995) 697–704.
[33] R. Pascanu, T. Mikolov, Y. Bengio, On the Difficulty of Training Recurrent Neural Networks. (2012).
[34] E. Choi, M.T. Bahadori, J.A. Kulas, et al., RETAIN: An Interpretable Predictive Model for Healthcare Using Reverse Time Attention Mechanism, (2016).
[35] M. Yuan, F. Li, X. Tian, et al., Risk factors for lung infection in stroke patients: a meta-analysis of observational studies, Expert Rev. Anti. Ther. 13 (2015) 1289–1298.
[36] S.E. Kasner, et al., Reliability and validity of estimating the NIH stroke scale score from medical records, Stroke 30 (1999) 1534–1537.